# A Treatise on Differential Equations of the Ordinary Kind

With an Analytical Lens

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## Contents

# 1 Compactness

We will now discuss the equivalence of topological and sequential compactness in the reals and in general metric spaces. Let us view the following theorem:

## 1.1 Limit Points

The limits of sequences ARE NOT necessarily limit points. Given some subset $E \subseteq \mathbb{R}$, $x$ is a limit point of $E$ if:

$$\forall \epsilon > 0 : B(x, \epsilon) \cap E \neq \varnothing$$

Notice how this only requires the intersection to be nonempty; which means there could possibly be a finite number of points in the intersection (but we will see this is not possible). Let us look at the following lemma:

**Lemma 1.1.** *$x$ is a limit point of $E$ if and only if*

$$\exists \{x_n\}_{n\in\mathbb{N}} \in (E \setminus \{x\})^{\mathbb{N}} : x_n \to x$$

*Proof.* $\implies$ : If $x$ is a limit point, then using the first countability of metric spaces, consider:

$$\epsilon_n \coloneqq \frac{1}{n+1}$$

Then by axiom of choice, choose $x_n \in B(x, \epsilon_n) \cap E \setminus \{x\}$, so that:

$$\forall n \in \mathbb{N} : |x_n - x| < \frac{1}{n+1} \implies x_n \to x$$

$\impliedby$ : Now for the converse. If there exists such a sequence $\{x_n\}_{n\in\mathbb{N}}$ such that:

$$\forall \epsilon > 0, \exists n_0 \geqslant 0, \forall n \geqslant n_0 : |x_n - x| < \epsilon$$

then we see that

$$\forall \epsilon > 0 : B(x, \epsilon) \cap E \setminus \{x\} \neq \varnothing$$

as $\forall n \geqslant n_0 : x_n \in B(x, \epsilon)$ and by definition of the sequence $x_n \in E \setminus \{x\}$. Thus proving equivalence. □

Now we introduce the following notion:

**Definition 1.1.** *An $\omega$-limit point of a set $E$ is one for which there exists a countably infinite number of points of $E$ within each punctured $\epsilon$-ball around $x$.*

It follows that every limit point in $\mathbb{R}$ is an $\omega$-limit point from Lemma 1.1, the sequential definition. This is because:

$$\forall \epsilon > 0, \exists n_0 \geqslant 0, \forall n \geqslant n_0 : x_n \in B(x, \epsilon) \cap E \setminus \{x\}$$

and the natural numbers $n_0$ and beyond are countably infinite (an injection of $\mathbb{N}$ into this set can easily be shown), thus $x$ is an $\omega$-accumulation point. The weakest sufficient condition is a $T_1$ space for such equivalence.

Observe the definition of sequential closure of some subset $E$:

**Definition 1.2.** *$\overline{E}$ is the closure of $E$ if and only if*

$$\forall \{x_n\}_{n\in\mathbb{N}} \in E^{\mathbb{N}} : x_n \to x \implies x \in \overline{E}$$

So by the equivalence stated in Lemma 1.1, $\{x_n\}_{n\in\mathbb{N}}$ belongs to $E^{\mathbb{N}}$ as well (it is not guaranteed that $x \in E$), but by sequential closure $x$ or the limit point must belong in $\overline{E}$.

Now steering our attention back to sequences, the limit of a sequence may not be a limit point. Consider an eventually constant or constant sequence. We see that it converges to whatever value it is eventually constant at; but there is an empty $\epsilon$-punctured ball intersection. So a limit point is a limit of a specific type of sequence, but not vice versa.

This distinction will be quite crucial in the following section.

## 1.2 Sequential Compactness

**Theorem 1.1** (Sequential Bolzano Weierstrass). *Any bounded sequence has a convergent subsequence.*

To prove this fact, we show first that any sequence in $\mathbb{R}$ has a monotonic subsequence.

*Proof.* Let us define the following set, we call "peaks" of a sequence:

$$P \coloneqq \{n \in \mathbb{N} : (\forall j > n : x_j < x_n)\}$$

In the case that $P$ is countably infinite, then we set:

$$n_0 \coloneqq \inf P \wedge \forall k > 0 : n_k \coloneqq \inf (P \setminus \{n_0, ..., n_{k-1}\})$$

In which case we are guaranteed that $\forall k \in \mathbb{N} : n_{k+1} > n_k$ which tells us that $x_{n_{k+1}} < x_{n_k}$, showing that there exists a strictly decreasing subsequence. In the case that $P$ is finite, we choose the following:

$$n_0 \coloneqq \sup(P) + 1$$

that way the peaks are fully ommited from the subsequence, as they will prove to be problematic in the construction. Then for any $k \in \mathbb{N}$, we are guaranteed that $n_k \notin P$ (since $\{n_k\}_{k\in\mathbb{N}}$ is strictly increasing and $n_0 \notin P$), which means:

$$\exists j > n_k : x_j \geqslant x_{n_k}$$

So we simple define:

$$\forall k \in \mathbb{N} : n_{k+1} \coloneqq \inf\{j > n_k : x_j \geqslant x_{n_k}\}$$

in which case we obtain a non-decreasing subsequence. Now any bounded sequence has a monotone subsequence, which converges. What this means is given some sequence $\{x_n\}_{n\in\mathbb{N}}$, we have that:

$$\exists y \in \mathbb{R}, \exists r > 0, \forall n \in \mathbb{N} : x_n \in B(y, r)$$

so the sequence has a convergent subsequence within the ball, but such a subsequence may not converge within the ball. This is due to the topological idea that the limit point certainly belongs in the closure of the set but not necessarily in the open set. Consider the bounded sequence:

$$\{x_n\}_{n\in\mathbb{N}} \in (0,1)^{\mathbb{N}} : \left(\forall n \in \mathbb{N} : x_n \coloneqq \frac{1}{n+2}\right)$$

Then $x_n \to 0$ yet $0 \notin (0, 1)$. This idea of closedness plays a key role in sequential compactness. $\square$

**Definition 1.3.** *A sequentially compact interval of $\mathbb{R}$ is one for which every sequence has a convergent subsequence that converges to a point within the interval.*

The previous Theorem 1.1 leads to a key theorem:

**Theorem 1.2.** *An interval of $\mathbb{R}$ is sequentially compact if and only if it is closed and bounded.*

$\Longleftarrow$ : This direction follows from Sequential Bolzano Weierstrass. If the interval is bounded, then any sequence has a convergent monotone subsequence. Since it is closed, the limit of the sequence, which is an adherent point, belongs in the closure.

$\Longrightarrow$ : Consider any convergent sequence $\{x_n\}_{n\in\mathbb{N}}$ such that $\forall n \in \mathbb{N} : x_n \in I$. Then by sequential compactness:

$$\exists\{n_k\}_{k\in\mathbb{N}} : x_{n_k} \to x \wedge x \in I$$

which forces $x_n \to x$. Thus all convergent sequences converge to points within the set (which by sequential closure) means $I$ is closed. Now assume the interval is not bounded, in which case:

$$\forall r > 0, \forall y \in \mathbb{R} : I \nsubseteq B(y, r)$$

Or equivalently (to bring about the sequential format, we use first countability):

$$\forall n \in \mathbb{N}, \forall y \in \mathbb{R}, \exists x \in I : x \notin B(y, n+1)$$

Choose some arbitrary $y \in \mathbb{R}$. Then $\forall n \in \mathbb{N}$ choose $x_n$ such that $x_n \notin B(y, n+1)$, so that:

$$\forall n \in \mathbb{N} : |x_n - y| > n + 1 > 1/2$$

Now assume there existed some $z$ such that some subsequence indexed by $\{n_k\}_{k\in\mathbb{N}}$ converged to $z$. Then:

$$\forall k \in \mathbb{N} : |x_{n_k} - y| \leqslant |x_{n_k} - z| + |y - z|$$

$$\Longrightarrow \forall k \in \mathbb{N} : ||x_{n_k} - y| - |y - z|| \leqslant |x_{n_k} - z|$$

Taking limit $k \to \infty$ we see that $n_k \to \infty$, thus:

$$||x_{n_k} - y| - |y - z|| \to 0$$

which is a contradiction because $|x_{n_k} - y| > n_k + 1$ which tells us that $|x_{n_k} - y| \to \infty$, so $x_{n_k} \nrightarrow z$ and since $z$ is arbitrary, this holds true for all such $z \in \mathbb{R}$.

**Lemma 1.2.** *For any set $I \subseteq \mathbb{R}$, $I$ is bounded if and only if it is totally bounded (ONLY for $\mathbb{R}$).*

*Proof.* $\Longleftarrow$ : If $I$ is bounded, then:

$$\exists r > 0, \exists y \in \mathbb{R} : I \subseteq B(y, r) \iff \exists r > 0, \exists y \in \mathbb{R}, \forall x \in I : x \in B(y, r)$$

We aim to show $I$ is totally bounded, which means:

$$\forall \epsilon > 0, \exists x_0, ..., x_m \in I : I \subseteq \bigcup_{i=0}^{m} B(x_i, \epsilon)$$

This converse direction is necessarily true; however, the forward direction is not necessarily true in all spaces. Consider the following:

$$M \coloneqq \max\{|x_i - x_j| : 0 \leqslant i, j \leqslant m\}$$

Then, choose some $x_i$; so without loss of generality, we choose $x_0$:

$$\forall x \in I, \exists x_i : x \in B(x_i, \epsilon) \implies |x - x_0| \leqslant |x - x_i| + |x_i - x_0| < \epsilon + M$$

$\Longrightarrow$ : Now to prove the forward direction. Assume $I$ is not totally bounded:

$$\exists \epsilon > 0, \forall x_0, ..., x_m \in I, \exists x \in I : x \notin \bigcup_{i=0}^{m} B(x_i, \epsilon)$$

then we aim to show:

$$\forall r > 0, \forall y \in \mathbb{R}, \exists x \in I : x \notin B(y, r)$$

$\square$

## 1.3 Limit Point Compactness